QUESTION 1

A) At least 25 different HTML elements

#	HTML Element	Purpose
1	html	Root of the HTML document
2	head	Contains document information
3	title	Browser page title
4	meta	Metadata
5	body	Visible page content
6	header	Header section
7	nav	Navigation
8	main	Main content
9	section	Content section
10	article	Independent content
11	h1	Main heading
12	h2	Section heading
13	h3	Subheading
14	p	Paragraph
15	strong	Important/bold text
16	a	Hyperlink
17	table	Table
18	caption	Table caption
19	thead	Table header
20	tbody	Table body
21	tr	Table row
22	th	Table heading
23	td	Table cell
24	ul	Unordered list
25	ol	Ordered list
26	li	List item
27	dl	Description list
28	dt	Description term
29	dd	Description definition
30	figure	Image/content figure
31	img	Image
32	figcaption	Figure caption
33	address	Contact information
34	form	Form
35	label	Form label
36	input	Form input
37	textarea	Multi-line input

38	button	Button
39	footer	Footer
40	br	Line break
41	hr	Horizontal rule

b) Semantic HTML structure

the structure is

```
<html>
|
├── <head>
|   ├── <meta>
|   └── <title>
|
└── <body>
    |
    ├── <header>
    |   ├── <h1>
    |   ├── <p>
    |   └── <nav>
    |
    ├── <main>
    |   |
    |   ├── <section> About Me
    |   |
    |   ├── <section> Education
    |   |
```

```
│   ├── <section> Skills
│   │
│   ├── <section> Projects
│   │
│   ├── <section> Services
│   │
│   ├── <section> Gallery
│   │
│   └── <section> Contact
│
└── <footer>
```

c) What type of website will you create and what content will it contain?

Type of Website and Content

I will create a **Personal Portfolio Website** using HTML. The purpose of the website is to introduce myself, showcase my education, skills, projects and services, and provide contact information.

The website will contain several clearly organized sections. These will include a **Home section** with my name and introduction, an **About Me section** describing myself, an **Education section** showing my academic background, and a **Skills section** listing my technical and personal skills.

It will also include a **Projects section** where I will showcase projects I have completed, a **Services section** describing services I can provide, and a **Gallery section** containing relevant images. Finally, there will be a **Contact section** containing my email address, telephone number and a contact form.

The website will use **semantic HTML elements** such as <header>, <nav>, <main>, <section>, <article>, <address> and <footer>. It will also demonstrate more than **25 different HTML elements** and more than **15 HTML attributes**.

No external CSS or JavaScript will be required because the main objective is to demonstrate my understanding of HTML elements, attributes and semantic structure.

Question 2: HTML Elements

1. **Which 5 elements did you find most challenging to implement and why?**

The five HTML elements I found most challenging to implement were

<table>, **<form>**, **<figure>**, **<nav>**, and **<fieldset>**.

a) **<table>**
The <table> element was challenging because I had to organize information into rows and columns. I used <table>, <tr>, <th>, and <td> to create my education table and also to organize parts of the website layout.
b) **<form>**
The <form> element was challenging because it contains several other elements such as <label>, <input>, <textarea>, and <button>. I had to make sure that each label was correctly connected to its corresponding input field.
c) **<figure>**
The <figure> element was challenging because I needed to correctly combine an image with its description using <figcaption>. I used it in the Gallery and Projects sections to display my images with appropriate captions.
d) **<nav>**
The <nav> element was challenging because I needed to create links that take the user to different sections of the same webpage. I used the href attribute together with section id attributes, for example:

2. **How did you use semantic elements to structure your content?**

I used semantic HTML elements to make the structure of my website clear and meaningful.

The **<header>** element was used for the top section of my website. It contains my name, portfolio title, introduction and navigation menu.

The **<nav>** element was used to contain the navigation links that allow visitors to move between different sections of the website.

The **<main>** element contains the main content of the webpage. Inside <main>, I created several <section> elements.

Each **<section>** represents a major part of my website, including:

- About Me
- Education
- Skills
- Projects
- Services
- Gallery
- Contact

I used **<article>** for individual pieces of independent content, such as my projects and the "Who Am I?" section.

The <**footer**> element was used at the bottom of the website to contain my name, copyright information and a "Back to Top" link.

3. Which element was most useful for organizing your layout and why?

   The <**table**> **element** was the most useful element for organizing my layout.

   I used tables not only for my education information but also to organize different areas of the website, such as the navigation, gallery and content sections.

   Example:

   <table width="90%" align="center" cellpadding="15">

QUESTION 3: HTML ATTRIBUTES

1) Which 3 attributes were essential for making your website functional?
The three attributes I found most essential were **href**, **src**, and **id**.

- **href attribute**

I used the href attribute in the navigation links to allow users to move between different sections of my website.

For example:
```
<a href="#about">About Me</a>
```

a) **src attribute**

I used the src attribute to tell the browser where to find my images.

For example:

```
<img
   src="images/profile.jpg"
   alt="Profile photograph of Lukumo Kibela"
>
```

b) **id attribute**

I used the id attribute to uniquely identify the different sections of my website.

For example:

```
<section id="about">
```

2) How did you use the class and id attributes differently?

In my current website, I mainly used the **id attribute**, while I did not need to use the class attribute.

The **id attribute** is used to give a unique identification to a particular element. For example:

```
<section id="about">
```

I gave each major section a unique ID:

- home
- about
- education
- skills
- projects
- services
- gallery
- contact

This allowed my navigation links to move directly to specific sections.

The **class attribute**, on the other hand, is normally used when I want to identify **multiple elements that belong to the same group**.

3) Which attribute helped improve user experience the most and why?

I used the alt attribute with my images, for example:

```
<img
   src="images/programming.jpg"
   alt="Computer programming and web development"
   width="350"
   height="250"
>
```

The alt attribute provides a description of an image.

It is useful because if an image fails to load, the alternative text can be displayed instead. It also helps people who use **screen readers** understand what an image represents.

Question 4: Development Process

1. How did you plan your website structure before coding?

   Before writing the HTML code, I first decided that I would create a **personal portfolio website**. I chose this type of website because it would allow me to demonstrate many different HTML elements and attributes while presenting information about myself.

   I planned the website by dividing it into clear sections. My initial structure was:

   Personal Portfolio Website

```
│
├── Header
│   ├── My Name
│   ├── Introduction
│   └── Navigation Menu
│
├── Main Content
│   ├── About Me
│   ├── Education
│   ├── Skills
│   ├── Projects
│   ├── Services
│   ├── Gallery
│   └── Contact
│
└── Footer
    ├── Copyright
    └── Back to Top
```

2. What was your approach to testing and debugging your HTML?

   I tested and debugged my HTML step by step rather than waiting until the entire website was completed.

   First, I saved my index.html file and opened it in a web browser. I checked whether the headings, paragraphs, navigation links, tables, images and contact form were displaying correctly.

   I then tested each navigation link individually. For example, when I clicked:

   ```
   <a href="#about">About Me</a>
   ```

3. What challenges did you face and how did you overcome them?

   One of the main challenges I faced was **organizing the webpage without using CSS**. Normally, CSS would be used to control the layout, spacing, alignment and appearance of a website.

   Since the assignment required me to demonstrate HTML, I used HTML elements and attributes such as `<table>`, `align`, `width`, `cellpadding`, and `cellspacing` to organize the content.

   Another challenge was **organizing the images correctly**. Initially, I had to make sure that the image files were placed in the correct folder. I solved this by creating an images folder inside my main project folder and using relative paths such as:

   ```
   <img src="images/project1.jpg" alt="Personal portfolio website project">
   ```

Question 5: Git & GitHub Implementation

1. What Git commands did you use during development?

During the development of my personal portfolio website, I used Git to track changes and manage different versions of my project. The main Git commands I used were:

- **git init** – Used to initialize Git in my project folder and create a new local repository.
- **git status** – Used to check which files were modified, added, or not yet tracked.
- **git add .** – Used to add all changed files to the staging area before committing.
- **git commit -m "message"** – Used to save a specific version of my project with a meaningful message.
- **git log** – Used to view my previous commits and see the development history of the project.
- **git branch** – Used to view or manage branches in the repository.
- **git remote add origin <repository-url>** – Used to connect my local Git repository to my GitHub repository.
- **git push -u origin main** – Used to upload my local project and commits to GitHub.
- **git pull origin main** – Used to download the latest changes from GitHub when necessary.

These commands helped me keep track of my work and maintain a history of changes made to the website.

2. How many commits did you make and what was your commit message strategy?

I made **5 commits** during the development of my personal portfolio website.

My commit messages were short, clear, and descriptive so that I could easily understand what was changed in each version. My commit strategy was to make a commit whenever I completed an important stage of development.

Example commit messages were:

1. **Initial commit - create portfolio website structure**
   Created the basic index.html file and project structure.
2. **Add personal portfolio content**
   Added my About Me, Education, Skills, Projects, and Services sections.
3. **Add images and gallery section**
   Added the images folder and included images in the website gallery.
4. **Add contact form and navigation**
   Added the navigation menu, contact section, form, and links between sections.
5. **Finalize HTML portfolio website**
   Made the final corrections, checked the HTML structure, and prepared the website for submission.

The commit messages followed a simple strategy: each message described the main change made in that version instead of using vague messages such as "changes" or "update".

3. Why is version control important for web development projects?

Version control is important because it allows developers to keep a history of changes made to a project. Git makes it possible to return to an earlier version if a new change causes a problem.

Version control is also useful because:

- It **tracks changes** made to files over time.
- It allows developers to **restore previous versions** of a project.
- It provides a **backup** of the project when it is uploaded to GitHub.
- It makes it easier for **multiple developers to work together** on the same project.
- It allows developers to work with **branches** when developing new features.
- It helps identify **who made changes and when** the changes were made.
- It provides a professional way to **manage and document the development process**.
- GitHub makes it easier to **share projects with lecturers, employers, or other developers**.

For my personal portfolio website, Git and GitHub helped me keep a record of my development process and ensured that I could safely save and access different versions of my website.

Question 6: Code Quality & Best Practices

1. How did you ensure your HTML was valid and error-free?

I ensured that my HTML was valid and error-free by carefully checking the structure of my code while developing the website. I opened the index.html file in a web browser regularly to check how the website was displayed.

I used the following methods:

- I checked that every opening HTML tag had the appropriate closing tag.
- I made sure that HTML elements were properly nested.
- I checked that attributes such as href, src, alt, and id were correctly written.
- I checked that navigation links matched the correct section IDs, for example, href="#about" links to id="about".
- I checked all image paths to ensure that the images in the images folder displayed correctly.
- I checked the spelling and capitalization of filenames because incorrect paths can prevent images from loading.
- I reviewed the page using the browser's developer tools to identify possible errors.
- I tested the navigation links, images, tables, and contact form after making changes.
- I removed unnecessary Markdown code fences from the HTML file because they are not valid HTML content.

I also reviewed the overall document structure to make sure it followed the correct HTML format, beginning with <!DOCTYPE html> and using appropriate <html>, <head>, and <body> elements.

2. What best practices did you follow for writing clean, readable code?

I followed several best practices to make my HTML code clean, organized, and easy to understand.

a. Used proper indentation

I indented nested elements so that the structure of the HTML was easy to follow. For example, content inside <main> was indented further than the <main> element itself.

b. Used semantic HTML

I used semantic elements such as:

- <header> for the page header
- <nav> for navigation
- <main> for the main content
- <section> for major content sections
- <article> for individual pieces of content
- <footer> for the footer

This makes the structure of the website clearer and improves accessibility.

c. Used meaningful headings

I organized the content using headings such as <h1>, <h2>, and <h3> in a logical order. This makes the website easier for users and screen readers to understand.

d. Used descriptive attributes

For images, I used the alt attribute to provide alternative descriptions. I also used meaningful id values such as about, skills, projects, and contact for navigation.

e. Organized files properly

I kept my main HTML file in the project root and placed website images inside a separate images folder:

Personal-Portfolio-Website/

- index.html
- images/
    - profile.jpg
    - programming.jpg
    - project1.jpg
    - project2.jpg

This made the project easier to manage.

f. Avoided unnecessary code

Because the assignment required an HTML-only website, I did not add external CSS or JavaScript. I focused on using HTML elements and attributes effectively to demonstrate my understanding of HTML.

g. Used comments where useful

Comments can be used to identify major sections of the code, making it easier to locate and edit specific parts of the website.

Overall, I tried to keep the code consistent, properly indented, logically organized, and easy for another person to read.

3. How would you improve your website if you had more time?

If I had more time, I would improve my website in several ways.

a. Add CSS styling

The biggest improvement would be adding CSS to make the website more visually attractive. I would add:

- A professional color scheme

- Better typography
- Responsive layouts
- Navigation styling
- Image sizing and spacing
- Cards for projects and services
- Better form styling
- Hover effects